

Felix Hoang

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Education

University of South Florida

Bachelor of Science in Computer Science, GPA: 3.6

Tampa, FL

Expected May 2028

Experience

Research Assistant

Sep 2025 - Present

Secure & Intelligent Systems Lab

- Automated the end-to-end research pipeline by engineering a data service with **mcp-*arxiv-server***, converting raw Arxiv PDFs into structured Markdown for standardized documentation.
- Orchestrated an **n8n** workflow to programmatically inject datasets and benchmarks as YAML metadata, enabling granular content-based search across a library of **200+** papers.
- Streamlined knowledge distribution by deploying a research portal via **Hugo** and **GitHub Actions**, reducing the publication cycle from several hours to **under 2 minutes**.

Tech Lead

Jun 2024 – Nov 2024

Summit Education

- Led a 4-member team to architect an e-waste ecosystem, placing in the **top 1% of 1,000+ national entries** Samsung Solve for Tomorrow hackathon by bridging IoT hardware with a scalable digital marketplace.
- Engineered a high-concurrency **Go** backend using **Goroutines** to process **500+** simultaneous inventory transactions with **sub-100ms** response times.
- Developed a **Python** edge layer on **Raspberry Pi** for real-time sensor processing and **Firebase** synchronization, achieving hardware-to-cloud latency of **200ms**.

Projects

End-to-End MLOps Platform for Real-Time Fraud Detection

- Architected a **FastAPI** inference service on **GKE** handling **2,000+ QPS** with a **P99 latency of 50ms**, using **Terraform** for dynamic resource scaling.
- Implemented a **GitOps** workflow with **ArgoCD** and **Helm** to automate model hot-swapping, reducing deployment lead time to **under 15 minutes** with **zero downtime**.
- Deployed a **Prometheus/Grafana** observability stack with automated data-validation gates to detect model drift and minimize false positives in high-traffic production environments.

Secure & Automated RKE2 Kubernetes Bootstrap | *Terraform, Ansible, Vault*

- Automated **High Availability (HA)** RKE2 cluster provisioning on **Proxmox**, reducing manual infrastructure setup from **4 hours to 15 minutes** using Ansible playbooks.
- Hardened cluster security by integrating **HashiCorp Vault** for OIDC authentication, replacing long-lived static credentials with dynamic **short-lived tokens**.
- Standardized multi-component deployments (Traefik, Longhorn) through a **single-config** YAML architecture, eliminating configuration drift across development and production nodes.

Home Lab & Applied Self-Hosting Projects | *ArgoCD, Kubernetes, Renovate*

- Managed a **3-node Proxmox** cluster hosting **20+ services** via **ArgoCD**, integrating **Renovate** for automated dependency updates and security patching across all Helm charts.
- Optimized resource allocation using **Descheduler** for pod rebalancing and implemented automated secrets rotation via the **External Secrets Operator**.

Technical Skills

DevOps & Infrastructure: Kubernetes, Docker, Terraform, Ansible, ArgoCD, Helm, GitHub Actions, Linux

Observability & Tools: Prometheus, Grafana, Loki, Tempo, Alloy, HashiCorp Vault, External Secrets Operator

Cloud & Virtualization: Google Cloud Platform (GCP), AWS, Proxmox VE, TrueNAS

Programming & Databases: Go (Golang), Python (FastAPI), Bash scripting, SQL, PostgreSQL (CloudNativePG)